

Better Than Guessing

Team Members: _____

You can fit a line to anything. The line does not care whether it makes sense. Today you will build the number that tells you whether a line was worth fitting — and then find a case where that number lies to you.

Open `Class18_Better_Than_Guessing_Skeleton.ipynb`. Answer here, on paper.

Part 1. Two Baselines

6 minutes

Someone reads you fifteen temperatures in random order and asks you to guess each one before it is revealed. You know nothing else.

1.1 What single number should you guess every time, and why that one rather than any other?

Now they tell you the cricket chirp rate before each guess. If chirps and temperature are related, you can do better than one fixed number — you can use a line.

That is the whole idea behind R^2 . It compares the error left over after using the line to the error you started with using only the mean:

$$R^2 = 1 - \frac{SS_{\text{Residual}}}{SS_{\text{Total}}}$$

1.2 Before computing anything: if the line were perfect, SS_{Residual} would be _____ and R^2 would be _____. If the line were no help at all, SS_{Residual} would equal _____ and R^2 would be _____.

Part 2. How Good Is the Cricket Line?

20 minutes

Fifteen observations of chirp rate and temperature. Work through the notebook in order.

2.1 The mean baseline. Record:

mean temperature _____ °F SS_{Total} _____

In one sentence, what do the red vertical segments in the plot represent?

2.2 Fit the line.

slope _____ intercept _____

Write the equation, and state what the slope means in plain English — with units.

2.3 Finish the calculation.

SS_{Residual} _____ R^2 _____ r _____ r^2 _____

Interpret the R^2 in a sentence a non-statistician would understand. What does the leftover $(1 - R^2)$ represent?

Last class you found $r = 0.80$ between two Pixar rating services and called it a strong correlation. Square it. Does “strong correlation” mean what you assumed it meant?

Part 3. Where the Line Stops Working

8 minutes

3.1 Use your equation to predict the temperature at each chirp rate:

19 chirps/sec → _____ °F

40 chirps/sec → _____ °F

And at 0 chirps/sec, the intercept says _____ °F

The observed chirp rates run from 14.4 to 20.0. Two of those three predictions are ones you should refuse to make. Which, and what is wrong with each?

3.2 The notebook keeps the fitted slope but forces the intercept to zero, and R^2 comes out around -14 . A negative R^2 means the line did worse than a flat guess at the mean. Look at what that line predicts at 15 chirps/sec compared to the real temperatures, and explain how a line can be worse than useless.

Part 4. A Better Number That Is a Worse Model

6 minutes

The notebook now fits a straight line to a second data set and reports its R^2 .

4.1 R^2 for the second data set _____ versus the crickets _____

Which fit has the higher R^2 ? Which data set is the straight line actually the right model for? Look at the plot before you answer.

4.2 Now compare the two residual plots. Sketch the shape of each, roughly:

crickets

second data set

4.3 So: does a high R^2 mean you have the right model? State what the residual plot told you that R^2 could not, in one or two sentences.